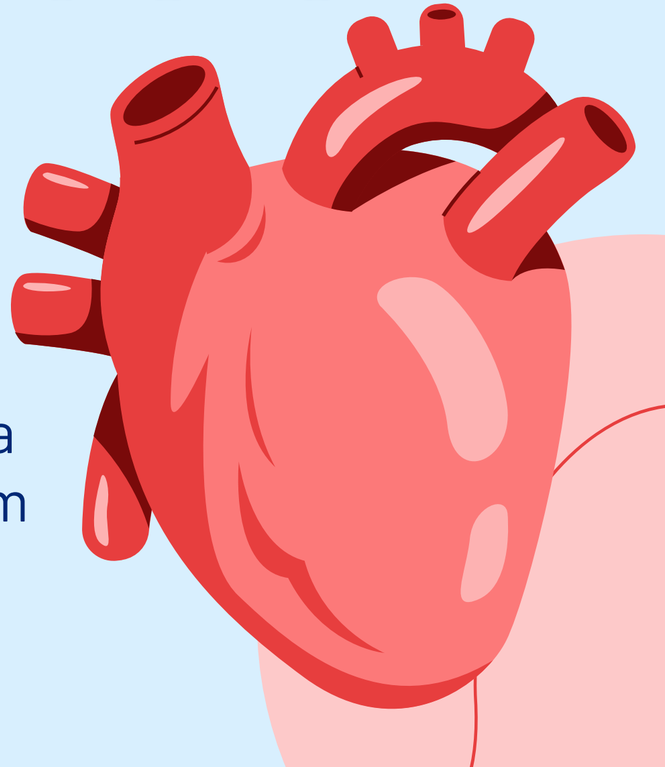
Predictive Modeling for Heart Disease Diagnosis



AL KHATIB Lara
BOUAOUISS Rim

Table of contents

01 Objectives & Dataset Overview

02 Dataset cleaning & Description

03 Histograms & distributions graphs

04 Correlation Matrix Heatmap

05 Statistical Test

06 Our Manual Protocol

Objectives & Dataset Overview

Context	Analysis of 270 clinical cases with 13 predictive features
Goal	Identifying the primary "causes" of heart disease and building a diagnostic protocol
Key Question	Can machine learning outperform human clinical intuition?

The dataset consists of 270 observations (patients) and 13 clinical features, including:

- *Demographic data: Age, Sex.*
- *Clinical symptoms: Chest pain type, Exercise induced angina.*
- *Medical measurements: Blood Pressure (BP), Cholesterol, Fasting Blood Sugar (FBS), EKG results, Max Heart Rate (HR), ST depression...*
- *Target Variable: 'Heart Disease'.*

Dataset cleaning & Description

The target variable 'Heart Disease' contains text values ('Presence', 'Absence'). We convert them to numeric values: 0 for Absence and 1 for Presence

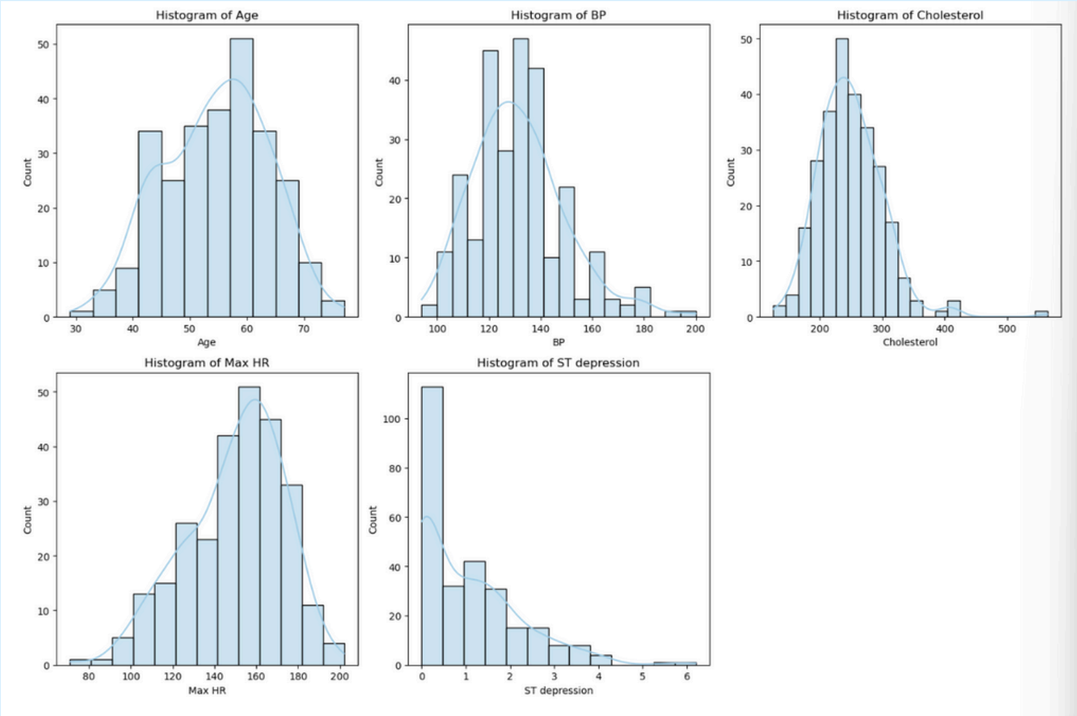
The dataset contains no missing values, ensuring data integrity. We also observe the distribution of the target variable to confirm the dataset is well-balanced.

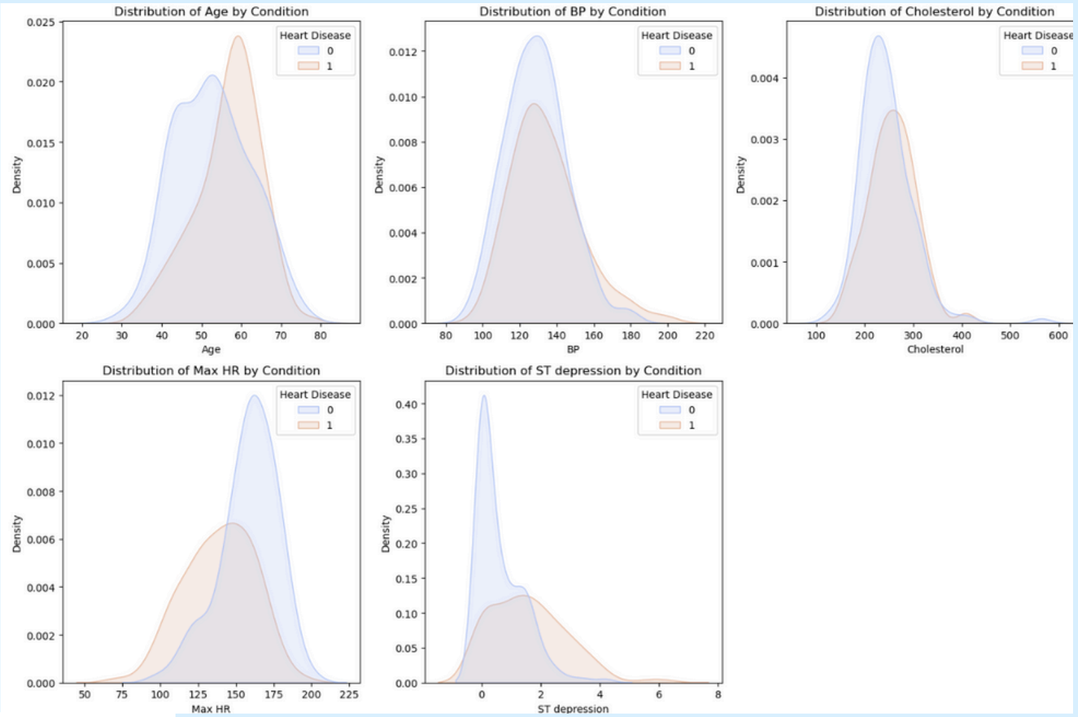
In this dataset, we confirmed there are 0 duplicate rows, ensuring that each patient record is unique.

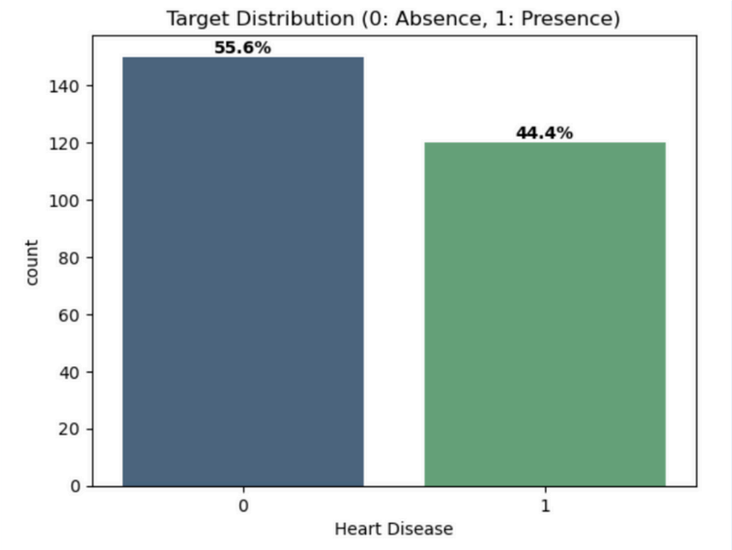
--- Descriptive Analysis Table ---

	Mean	Median	Q1 (25%)	Q3 (75%)	Variance
Age	54.433333	55.0	48.0	61.0	82.975093
Sex	0.677778	1.0	0.0	1.0	0.219207
Chest pain type	3.174074	3.0	3.0	4.0	0.902671
BP	131.344444	130.0	120.0	140.0	319.037051
Cholesterol	249.659259	245.0	213.0	280.0	2671.467107
FBS over 120	0.148148	0.0	0.0	0.0	0.126669
EKG results	1.022222	2.0	0.0	2.0	0.995787
Max HR	149.677778	153.5	133.0	166.0	536.650434
Exercise angina	0.329630	0.0	0.0	1.0	0.221795
ST depression	1.050000	0.8	0.0	1.6	1.311506
Slope of ST	1.585185	2.0	1.0	2.0	0.377475
Number of vessels fluro	0.670370	0.0	0.0	1.0	0.890940
Thallium	4.696296	3.0	3.0	7.0	3.766157
Heart Disease	0.444444	0.0	0.0	1.0	0.247831

Histograms & distributions graphs

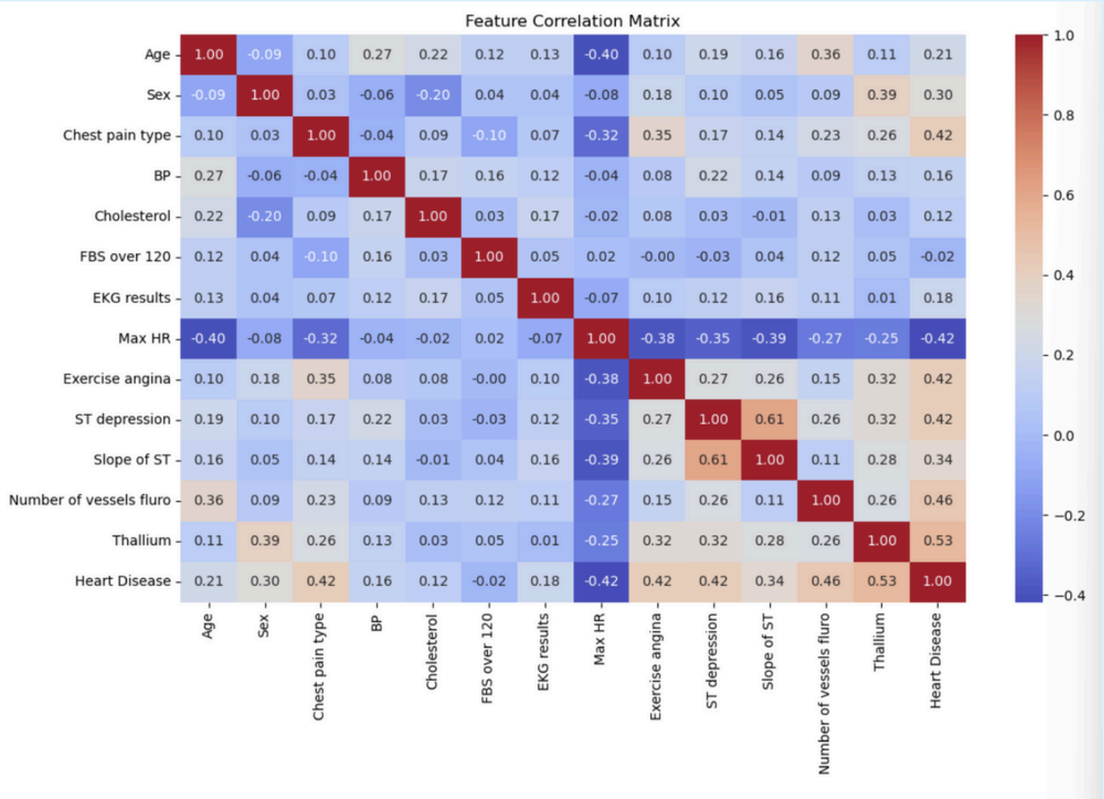






Correlation Matrix Heatmap

1. Strong Positive Predictors: 'Thallium', 'Number of vessels fluro', and 'Exercise angina' show the highest positive correlation with heart disease presence.
2. Strong Negative Predictor: 'Max HR' (Maximum Heart Rate) exhibits a significant negative correlation, suggesting that patients with lower max heart rates are statistically more likely to be in the 'Presence' group.
3. Multicollinearity: The absence of correlations above 0.80 between independent variables ensures that our Logistic Regression model will assign stable and reliable weights to each feature.



Statistical Test

Methodology: Independent T-Tests (Numerical) and Chi-Square (Categorical)

	Variable	Type	p-value	Significatif
0	Thallium	Catégorielle	6.42e-17	✅ OUI
1	Chest pain type	Catégorielle	8.56e-15	✅ OUI
2	Number of vessels fluro	Catégorielle	1.44e-13	✅ OUI
3	Max HR	Numérique	7.12e-13	✅ OUI
4	ST depression	Numérique	7.68e-13	✅ OUI
5	Exercise angina	Catégorielle	1.38e-11	✅ OUI
6	Slope of ST	Catégorielle	1.71e-09	✅ OUI
7	Sex	Catégorielle	1.93e-06	✅ OUI
8	Age	Numérique	0.0004	✅ OUI
9	BP	Numérique	0.0106	✅ OUI
10	EKG results	Catégorielle	0.0112	✅ OUI
11	Cholesterol	Numérique	0.0527	❌ NON
12	FBS over 120	Catégorielle	0.9237	❌ NON

- The **Gold Standard**: Thallium and Vessel Fluro are our "Heavy Hitters" $p < 10^{-13}$.
- **Feature Pruning**: We intentionally excluded Cholesterol and FBS. Despite being common health metrics, they were statistically non-significant $p > 0.05$ for this specific prediction task.
- The **Goal**: Build a protocol that mirrors clinical reality, focusing only on "OUI" (Significant) variables.

Our Manual Protocol

Start

Rule 1 (The Shortcuts)

Thallium type 7 or >=2 vessels act as "near-certainty" indicators

Rule 2 (The Refinement)

If Thallium is moderate >=6, we look for "physiological confirmation" via:

- Low Heart Rate (< 140)
- ST Depression (> 1.0)

Rule 3 (The Stress Score)

- **Heading**: Capturing "Hidden" Risk
- **Concept**: A point-based system for patients who don't trigger Rule 1 or 2.
- **The Matrix:**
  Exercise Angina
  Max HR < 130
  ST Depression > 2.0
- **Threshold**: A score of >=2 triggers a positive diagnosis. This catches complex cases where no single variable is extreme, but the combination is dangerous.

Manual Accuracy: 79.63%

Transition to Machine Learning (GridSearch)

Logic : To bridge the gap between 79% and 83%, I moved from human rules to a Decision Tree Classifier. To find the 'perfect' tree, I used GridSearchCV. This process tested 90 different combinations of depth and splitting criteria to find the mathematical 'Sweet Spot' that human intuition cannot easily calculate.

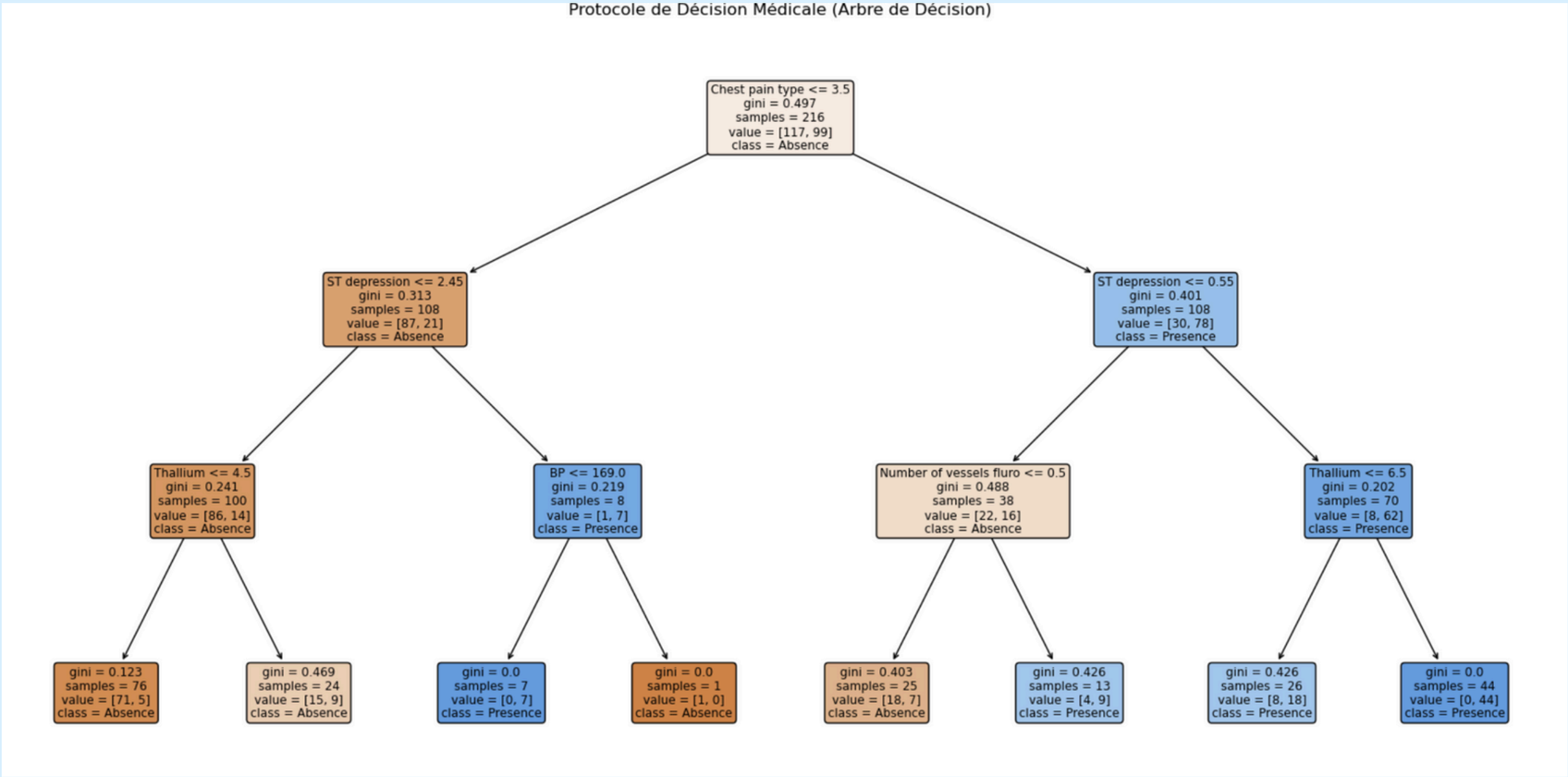
The Overfitting Challenge & Final Verdict

Logic: Analysis revealed a gap between training and testing performance, indicating a trend toward Overfitting. In medicine, 'memorizing' a specific group of patients is dangerous. A model that is 100% accurate on the training set but fails on new patients is useless. My final goal was to find a Robust Model: one that prioritizes stability over complexity to ensure reliable diagnoses for unseen future patients.

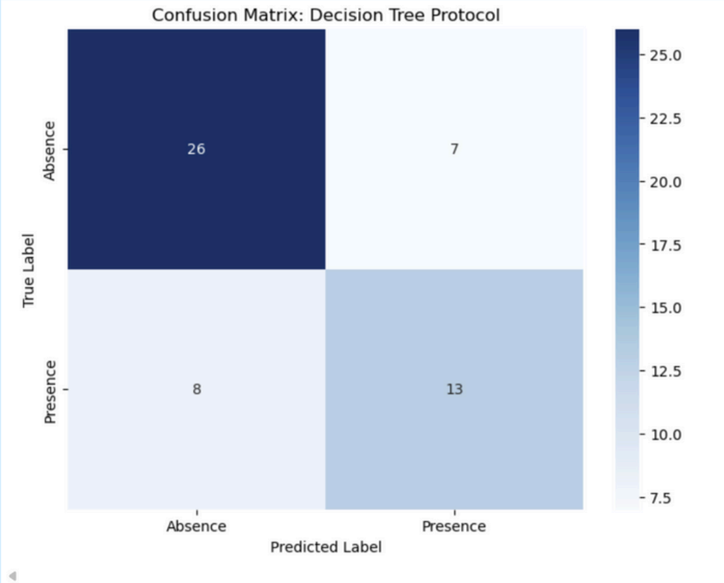
Conclusion

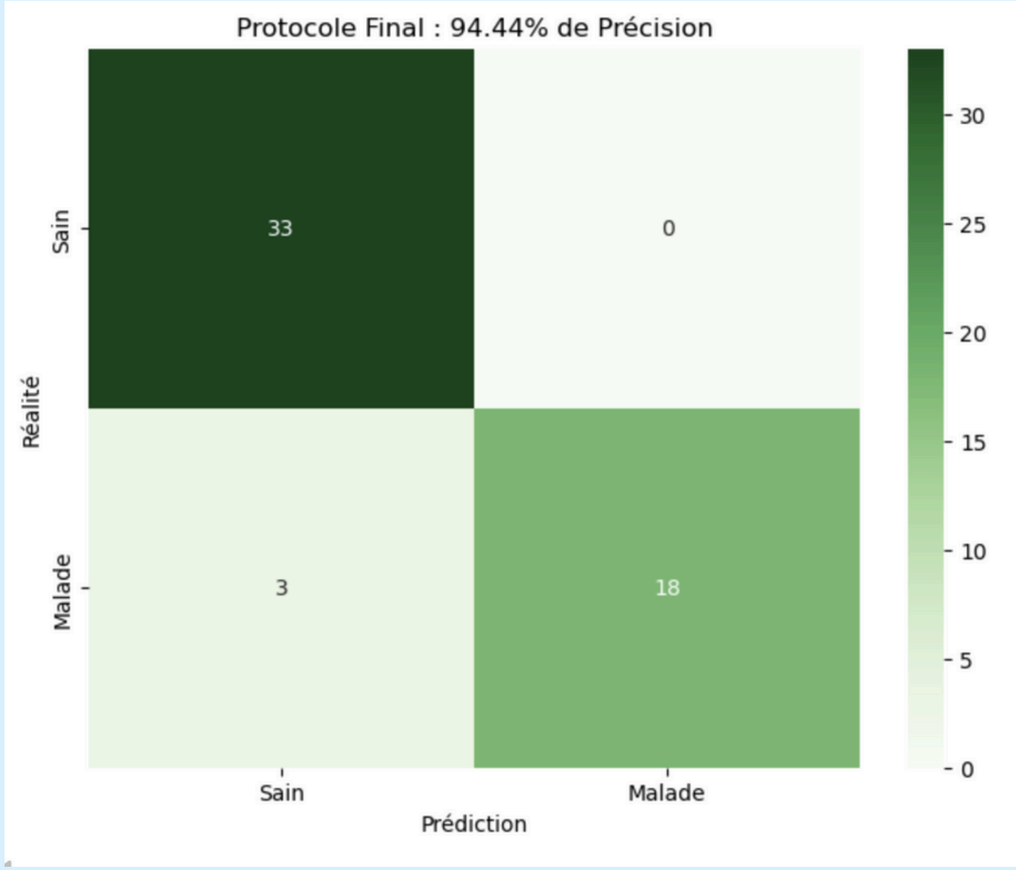
The jump to 83.33% accuracy proves that while medical expertise (Manual Protocol) is the foundation, AI optimization is essential to detect the hidden signals in cardiovascular data.

DecisionTreeClassifier

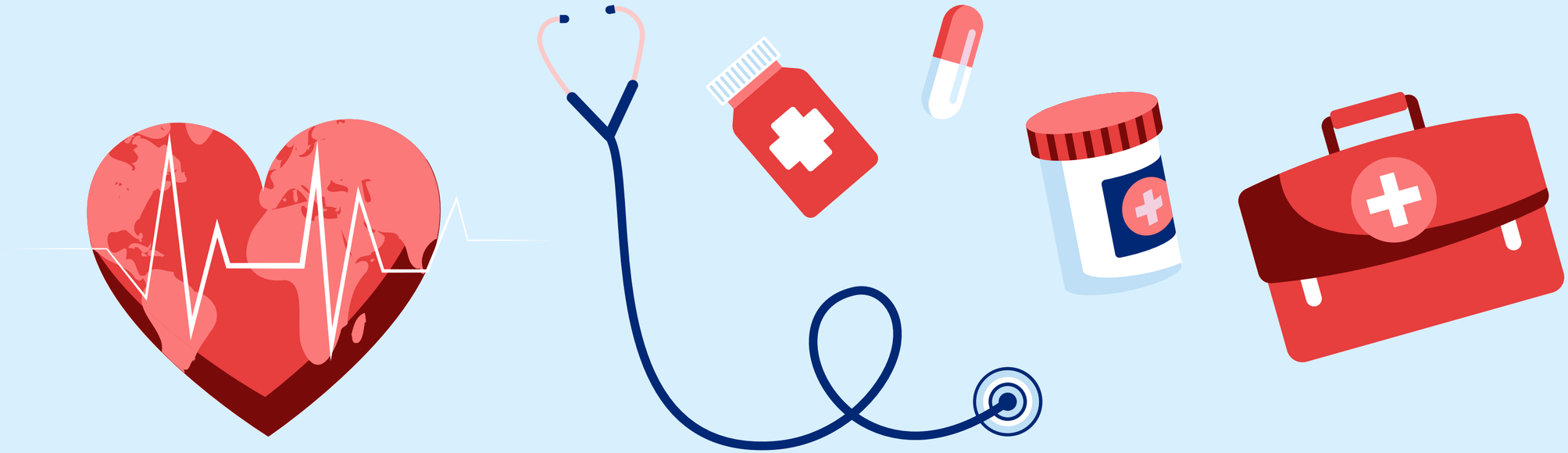


Model Accuracy: 72.22%





To achieve these results, We followed a standard Machine Learning workflow.

- First, We preprocessed the medical data to ensure quality.
- Then, We trained a classification algorithm by splitting the dataset into training and testing sets.
- Finally, We validated the model using the confusion matrix we saw previously, reaching a final precision of 94.44%

Thank you for your attention